

Unit V: Fourier Transforms

The Fourier cosine transform of a real function is the real part of the full complex Fourier transform

RECALL

If $f(t)$ be defined in $0 \leq t < \infty$ and $f(t)$ is extended as even function in $(-\infty, \infty)$ satisfying the condition on Fourier integral theorem. The Fourier cosine transform for the function $f(t)$ is defined by

$$F_C(\omega) = \int_0^{\infty} f(t) \cos \omega t \, dt$$

Where ' ω ' is a parameter

The Fourier sine and cosine transforms are forms of the Fourier transform that do not use complex numbers or require negative frequency. They are the forms originally used by Joseph Fourier and are still preferred in some applications, such as signal processing or statistics

DEFINITION

INVERSE FOURIER TRANSFORM

If $F_C(\omega)$ is the Fourier cosine transform of $f(t)$ which satisfies Dirichlet's condition in every finite interval $(0, l)$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_C(\omega) \cos \omega t \, d\omega$$

$f(t)$ is called inverse Fourier cosine transform of $F_C(\omega)$

1. Find the inverse Fourier cosine transform of $e^{-\omega}$

$$F_C(\omega) = e^{-\omega}$$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_C(\omega) \cos \omega t \, d\omega$$

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$$f(t) = \frac{2}{\pi} \int_0^{\infty} e^{-\omega} \cos \omega t \, d\omega$$

$$\text{w.k.t } \int_0^{\infty} e^{-a\omega} \cos \omega t \, d\omega = \frac{a}{a^2 + t^2} \text{ and } \int_0^{\infty} e^{-a\omega} \sin \omega t \, d\omega = \frac{t}{a^2 + t^2}$$

$$f(t) = \frac{2}{\pi} \left(\frac{1}{1^2 + t^2} \right)$$

2. Find $f(t)$ if its Fourier cosine transform is $\mathcal{F}_C(\omega) = \begin{cases} k - \frac{\omega}{2} & \text{for } \omega < 2k \\ 0 & \text{for } \omega > 2k \end{cases}$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_C(\omega) \cos \omega t \, d\omega$$

$$f(t) = \frac{2}{\pi} \int_0^{2k} \left(k - \frac{\omega}{2} \right) \cos \omega t \, d\omega + \int_{2k}^{\infty} (0) \cos \omega t \, d\omega$$

Integration by parts

$$f(t) = \frac{2}{\pi} \left[\left(k - \frac{\omega}{2} \right) \frac{\sin \omega t}{t} \right]_{\omega=0}^{\omega=2k} - \left[\left(-\frac{1}{2} \right) \frac{-\cos \omega t}{t^2} \right]_{\omega=0}^{\omega=2k}$$

$$f(t) = \frac{2}{\pi} \left[\left[\left(k - \frac{2k}{2} \right) \frac{\sin 2kt}{t} - \left(k - \frac{0}{2} \right) \frac{\sin 0}{t} \right] - \frac{1}{2} \left[\frac{\cos 2kt}{t^2} - \frac{\cos 0}{t^2} \right] \right]$$

$$f(t) = \frac{2}{\pi} \left[\frac{-1}{2} \left[\frac{\cos 2kt}{t^2} - \frac{1}{t^2} \right] \right]$$

$$f(t) = \frac{1}{\pi} \left[\frac{\sin^2 kt}{t^2} \right]$$

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